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Problem 3.1 The reduced row echelon form of the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 6 & 2 & -4 & -8 \\ 0 & 0 & 3 & 1 & -2 & -4 \\ 2 & -3 & 1 & 4 & -7 & 1 \\ 6 & -9 & 0 & 11 & -19 & 3 \end{pmatrix}$$

is the matrix

$$\mathbf{rref}(\mathbf{A}) = \begin{pmatrix} 1 & -3/2 & 0 & 11/6 & -19/6 & 0 \\ 0 & 0 & 1 & 1/3 & -2/3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Find all solutions to the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$.

Definition 3.2 The set of all solutions to a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ is called the **nullspace of matrix $\mathbf{A}$** , and is denoted $\mathbf{NS}(\mathbf{A})$.

Solution: First note that each variable corresponds to a column in the reduced row echelon matrix

$$\begin{matrix} & \textcolor{brown}{x} & \textcolor{blue}{y} & \textcolor{brown}{z} & \textcolor{blue}{u} & \textcolor{blue}{v} & \textcolor{brown}{w} \\ \begin{pmatrix} \textcolor{brown}{1} & -3/2 & 0 & 11/6 & -19/6 & 0 \\ 0 & 0 & \textcolor{brown}{1} & 1/3 & -2/3 & 0 \\ 0 & 0 & 0 & 0 & 0 & \textcolor{brown}{1} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

Any column that contains a pivot is called a **pivot column**. A variable whose corresponding column is a pivot column is called a **dependent variable** or **pivot variable**. The other variables are called **free variables**. In this problem, $\textcolor{brown}{x}$, $\textcolor{brown}{z}$, $\textcolor{brown}{w}$ are **dependent variables**, and $\textcolor{blue}{y}$, $\textcolor{blue}{u}$, $\textcolor{blue}{v}$ are **free variables**.

Let $\mathbf{x} = \begin{pmatrix} \textcolor{brown}{x} \\ \textcolor{blue}{y} \\ \textcolor{brown}{z} \\ \textcolor{blue}{u} \\ \textcolor{blue}{v} \\ \textcolor{brown}{w} \end{pmatrix}$. Then the system $\mathbf{Ax} = \mathbf{0}$ is equivalent to the system of equations

$$\textcolor{brown}{x} - \frac{3}{2}\textcolor{blue}{y} + \frac{11}{6}\textcolor{blue}{u} - \frac{19}{6}\textcolor{blue}{v} = 0$$

$$\textcolor{brown}{z} + \frac{1}{3}\textcolor{blue}{u} - \frac{2}{3}\textcolor{blue}{v} = 0$$

$$\textcolor{brown}{w} = 0$$

$$0 = 0.$$

The last equation gives us no information so we discard it.

$$\textcolor{brown}{x} - \frac{3}{2}\textcolor{blue}{y} + \frac{11}{6}\textcolor{blue}{u} - \frac{19}{6}\textcolor{blue}{v} = 0$$

$$z + \frac{1}{3}u - \frac{2}{3}v = 0$$

$$w = 0$$

Note that because we used the reduced row echelon form to solve this system, each equation directly expressed one pivot variable as a linear combination of the free variables. Start by assigning parameters to each of the free variables.

$$y = c_1 \quad \text{for a parameter } c_1,$$

$$u = c_2 \quad \text{for a parameter } c_2,$$

$$v = c_3 \quad \text{for a parameter } c_3.$$

Next express each pivot variable in terms of the parameters.

$$x = \frac{3}{2}y - \frac{11}{6}u + \frac{19}{6}v = \frac{3}{2}c_1 - \frac{11}{6}c_2 + \frac{19}{6}c_3$$

$$z = -\frac{1}{3}u + \frac{2}{3}v = -\frac{1}{3}c_2 + \frac{2}{3}c_3$$

$$w = 0$$

Therefore the general solution is given by

$$\begin{aligned} \begin{pmatrix} x \\ y \\ z \\ u \\ v \\ w \end{pmatrix} &= \begin{pmatrix} \frac{3}{2}c_1 - \frac{11}{6}c_2 + \frac{19}{6}c_3 \\ c_1 \\ -\frac{1}{3}c_2 + \frac{2}{3}c_3 \\ c_2 \\ c_3 \\ 0 \end{pmatrix} = \begin{pmatrix} (3/2)c_1 \\ c_1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} (-11/6)c_2 \\ 0 \\ (-1/3)c_2 \\ c_2 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} (19/6)c_3 \\ 0 \\ (2/3)c_3 \\ 0 \\ c_3 \\ 0 \end{pmatrix} \\ &= c_1 \begin{pmatrix} 3/2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -11/6 \\ 0 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 19/6 \\ 0 \\ 2/3 \\ 0 \\ 1 \\ 0 \end{pmatrix}. \end{aligned}$$

In particular, each of the three vectors

$$\begin{pmatrix} 3/2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -11/6 \\ 0 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 19/6 \\ 0 \\ 2/3 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

is a solution to the original homogeneous equation. And the set of all solutions is found by taking all linear combinations of these solutions.

3. Worked example: solving a homogeneous linear system

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